

A novel laser stripe key point tracker based on self-supervised learning and improved KCF for robotic welding seam tracking

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ABSTRACT

Laser vision based real-time welding seam tracking has emerged as a potent strategy for enabling intelligent robotic welding. And trackers based seam key point tracking algorithms demonstrate remarkable adaptability to complex welding environments. This paper proposed a self-supervised robust KCF (Kernelized Correlation Filter) tracker for seam key point tracking, which could be a novel approach to achieve autonomous seam tracking. Firstly, a self-supervised global-local feature extraction network is constructed, which can guide the model to focus on both global semantic and local texture features of laser stripes, thereby establishing a solid groundwork for stable key point tracking. Subsequently, a robust KCF tracking algorithm is presented. A multi-template enhanced tracker generation strategy is designed, and the corresponding analytical solution is derived, which can improve the tracker's representation capability of stripe features without significantly increasing computational complexity. Experimental results demonstrate that compared to traditional algorithms, the proposed algorithm exhibits advantages in tracking accuracy, stability, and real-time performance. Moreover, since the algorithm minimally relies on manually labeled data, it holds promise as a technological means to achieve fully autonomous seam tracking in actual welding production.

1. Introduction

Welding, as an essential component of modern manufacturing, is extensively employed in industries such as shipbuilding, aerospace, and construction [1]. With the escalating emphasis on welding precision and production efficiency, the intelligentization of welding processes centered around welding robots has become an inevitable trend in the development of welding manufacturing [2]. Nonetheless, traditional robotic systems predominantly hinge on teach-and-playback methodologies for welding tasks, coupled with manual verification, thereby engendering a decline in both production efficacy and welding robustness. Attaining autonomous welding capabilities in robots has thus emerged as a pivotal focal point in the paradigm shift towards intelligent welding manufacturing [3].

In recent years, welding robots have benefited from innovative sensor technologies and intelligent algorithms, leading to significant advancements in the level of welding intelligence [4]. Among these advancements, laser vision-based seam tracking technology is considered one of the effective means to achieve autonomous robotic welding [5]. However, the complex and dynamic nature of welding processes

presents significant challenges for accurately and stably extracting visual features of welding seams. Researchers worldwide have conducted extensive research on this topic.

Laser vision sensors are employed to precisely track the welding seam by extracting key points from the laser stripe patterns. Presently, predominant methodologies can be roughly categorized into two types based on their implementation principles: rule-based algorithms for key point extraction and tracker-based algorithms for key point tracking. Each is elaborated as follows:

Rule-based key point extraction algorithms extract key points of weld seams based on predefined rules or criteria derived from the properties of the weld seams. These algorithms typically involve techniques such as grayscale analysis, edge detection, morphological analysis, and geometric modeling to identify the positions of key points. Fang et al. [6] employed the grayscale centroid method to extract stripes, subsequently fitting the laser stripe equation to obtain key points of fillet weld seams. Shao et al. [7] utilized adaptive threshold to segment stripes and extracted key points of narrow gap butt weld seams based on morphological features of stripes. To augment the versatility of the algorithm across different types of weld seams, Wang et al. [8] based

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on LoG filtering to extract stripe positions, then leveraged the RANSAC algorithm for fitting and determining joint types; thereby facilitating tracking of various seam types such as fillet, lap, and butt joints. Li et al. [9] abstracted various types of weld seams into corresponding string structures; combined with stripe extracted via grayscale centroids, then achieved recognition and tracking of multiple typical joints.

In the aforementioned methods, stripes are extracted solely based on traditional features such as grayscale and edges, making the extraction susceptible to interference from welding noises such as smoke and arc light. In response to this challenge, scholars have applied deep learning algorithms for laser stripe feature extraction: Yang et al. [10] proposed a residual bidirectional ConvLSTM structure to remove welding noise from images; Wu et al. [11] designed an LSEGN for stripe segmentation, while Zhao et al. [12] trained an ERFNet to directly extract key points of weld seams. Ma et al. [13] regarded the extraction of weld seam key points as an object detection task, and proposed a key points extraction algorithm for complex weld seam based on Shuffle-YOLO, enabling the extraction of key points for butt joints, lap joints, and irregular joints. Additionally, Zhao et al. [14] introduced CGAN to reduce welding noise and employed the Yolov5 algorithm to detect key points of weld seams, which further improved the stability of the tracking process. Overall, these deep learning-based approaches demonstrate good performance under the interference of complex welding noise, significantly broadening the applicability of algorithms to different types of weld seams.

In contrast to rule-based methods, Tracking-based approaches extract key points of weld seams before welding. And then a filter is computed based on the features around the key points through regression. Finally, this filter is utilized to search for the positions most similar to the key points in subsequent images. Multiple tracking methods are employed in the tracking of seam key points, such as KCF [15,22], STC [16], ECO [17–19], CCOT [20], SiamFC [21], etc. Since trackers are generated online, this method naturally exhibits good adaptability to various types of weld seams.

However, under the interference of welding noise, it is difficult for tracking algorithms to ensure the absolute accuracy of key points' coordinates, leading to some fluctuation in the detected key point sequences. To address this issue, some scholars employ denoising methods to improve image quality, such as GAN [22], C-GAN [19], etc.; or establish state space equations for key points and utilize filtering algorithm such as Kalman filtering [16], and particle filtering [17], to reduce random fluctuations of key points.

While these methods stabilize the tracking process via image pre-processing or trajectory smoothing, they do not inherently enhance tracker capability. Since the aforementioned trackers predominantly employ traditional features such as HOG and CN, which have limited ability to discriminate laser stripe from welding noise. Consequently, these trackers are not entirely suitable for welding seam tracking scenarios. Zou et al. [18] designed a Unet network to segment stripes and used its deep features to enhance the stability of the ECO tracking algorithm. Xu et al. [22] introduced a generative adversarial network supervised by tracking features to enhance the accuracy of the DCF tracking algorithm. Xiao et al. [21] specifically developed LSFEnet for stripe tracking tasks, thereby enhancing the effectiveness of the SiamFC algorithm in the process of welding seam tracking. These tracking methods combine trackers with features extracted from deep neural networks, significantly improving the stability of the tracker and its robustness to noise by leveraging the representation capabilities of network features for stripes.

Summarily, the random fluctuations of welding noises during the welding process impose significant demands on the robustness of weld seam tracking algorithms. Existing algorithms have effectively addressed the issue through various means, and successfully extracted visual features of weld seams. Rule-based approaches aim to accurately locate key points that represent critical positions along the weld seam, while tracking-based algorithms focus on continuously monitoring and updating the positions of key points during the welding process. Both

types of methods play crucial roles in achieving precise and reliable welding trajectory tracking control.

Compared to traditionally manually designed features, deep features extracted by neural networks exhibit stronger resistance to interference. In addressing welding noise, deep learning-based algorithms demonstrate higher adaptability and robustness, thus enabling more stable capturing of weld seam features in complex welding scenarios. However, these algorithms are fundamentally data-driven, necessitating a substantial number of labeled samples for training. Limited samples may fail to cover all situations in complex welding scenarios, thereby diminishing the model's generalization ability.

Aiming at this challenge, this paper presents a self-supervised robust KCF laser stripe key point tracker for robotic welding seam tracking. The main contributions consist of two parts. Firstly, from the perspective of model structure and training strategy, a laser stripe feature extraction network is designed. By optimizing the network structure to enhance the model's feature extraction capability and employing self-supervised learning to guide the network's focus on the common features of laser images, a foundation is laid for stable tracking of stripes' key points. Secondly, an robust KCF tracker that improved by multi-template strategy is proposed to enhance the discriminations of the tracker to stripes, ensuring tracking stability under complex welding noise interference.

The remaining parts of this paper are arranged as follows: Sect. 2 introduces the experimental system, Sect. 3 elaborates on the laser stripe feature extraction network based on self-supervised learning. Sect. 4 proposes a multi-template improved KCF key point tracking algorithm. Sect. 5 presents experiments to demonstrate the accuracy and advantages of the algorithm. The conclusion is provided in Sect. 6.

2. Self-supervised robust KCF tracker based weld seam tracking system

The schematic diagram of the welding seam tracking system used in this paper is depicted in Fig. 1, comprising three main components: the industrial personal computer (IPC), Laser vision sensor (LVS), and the FANUC robot.

The IPC runs the weld seam tracking software, which controls the LVS to acquire laser stripe images during the welding process. The software then processes these images to extract key points of the weld seam while simultaneously recording the corresponding robot coordinates. Using the measurement model of the LVS, the key points are transformed into 3D coordinates in the robot's base coordinate system. Finally, these coordinates are stored in a buffer and await transmission to the robot for execution.

On the robot side, a TCP/IP communication protocol connects it to the weld seam tracking software. Throughout the tracking process, the robot continuously requests the trajectory point from the software to achieve real-time correction of the welding trajectory. To ensure trajectory accuracy and tracking stability, a FIFO (First In First Out) queue of trajectory points is established, recording the coordinates of weld seam key points from the current position of the welding torch to the

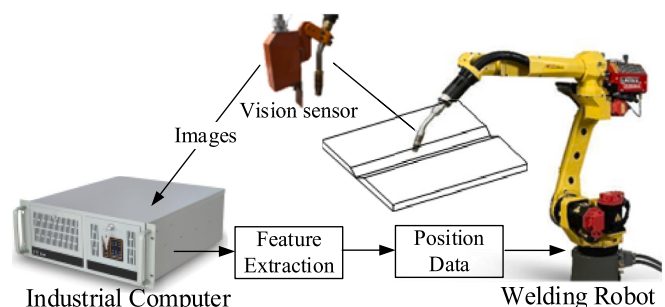


Fig. 1. Schematic diagram of the welding seam tracking system.

laser line measurement position. These points are then fed back to the robot through coordinate interpolation with a fixed step size.

The visual sensor comprises an industrial CCD camera and a line laser, supplemented by a narrow-band filter to mitigate interference from welding arc. LVS is installed at the front end of the robot's welding torch in an eye-in-hand configuration, following the measurement model as depicted in Eq. 1.

$$P_b = {}^bT_c^T TP_l \quad (1)$$

Here, P_b represents the homogeneous coordinates of the welding seam key points in the robot's base coordinate system, while P_l denotes the homogeneous coordinates of the key points in the image coordinate system obtained through image processing. bT_c stands for the transformation matrix from the tool coordinate system to the robot's base coordinate system, which can be obtained in real-time through communication with the robot. lT_c represents the intrinsic matrix of the LVS, which transforms image coordinates into three-dimensional coordinates in the sensor coordinate system. cT signifies the hand-eye transformation matrix between the LVS and the welding torch tip, which can be solved using the classical Tsai algorithm. The transformation matrices are obtained using the calibration method from our previous work [23].

3. Self-supervised learning based feature extraction network

The process of the proposed algorithm is illustrated in Fig. 2. Initially, a neural network is employed to extract features from both the target and the image to be detected simultaneously. Subsequently, target's features are utilized to train a tracker online. Finally, the tracker is used to search a position on the image to detected based on correlation operations, which is the position of the key point. This section will primarily elucidate the composition of the network model, while the training method for the tracker will be discussed in the Sect. 4.

As previously stated, the stronger the representation capability of features used in key points tracking process of stripes, the better the tracking performance. Therefore, this paper initiates from the model architecture and training strategy to design a network for extracting stripe features, thereby laying the foundation for the stability of the tracking process.

3.1. Self-supervised global-local feature extraction network

Previous research has demonstrated that local information contributes to precise localization but may encounter challenges in handling noise interference, whereas global information enhances noise resistance yet often lacks intricate details. Convolutional Neural Networks (CNNs), serving as local feature extraction networks, excel in capturing high-frequency features like textures and sharp edges. Nevertheless, they may encounter difficulties in grasping global semantic information. To ensure robust feature representation and enhance noise resilience, CNNs frequently necessitate techniques such as enlarging the convolutional kernel or stacking convolutional layers to expand the model's receptive field. Conversely, the latest Transformer architectures boast formidable capabilities in establishing long-range dependencies, facilitating the assimilation of global information. Therefore, in this paper, we establish a Self-supervised global-local feature extraction network

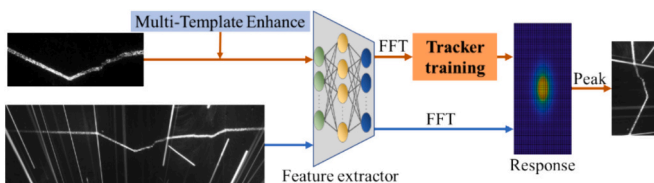


Fig. 2. The flowchart of the proposed algorithm.

(SSGLNet) by amalgamating CNN and Transformer architectures to fortify the model's feature extraction prowess, as shown in Fig. 3.

The architecture of designed network, as illustrated in the diagram, consists of two main parts: the feature extraction block and the feature fusion block. The feature extraction block adopts a residual structure, where the input feature maps are added to the output to ensure model performance. And the feature fusion block is utilized to merge the feature maps from different levels and enhance the model's representational capacity. The detailed descriptions of each block's structure are presented below:

3.1.1. Feature extraction block

The feature extraction block follows the design of the Inception structure [24], as shown in Fig. 4, comprising a CNN branch and a Transformer branch, which are respectively employed to extract local and global information.

The convolutional branch utilizes convolutional kernels of sizes 3×3 , 5×5 , and 7×7 to extract features with different receptive fields. Additionally, depthwise separable convolution [25] is employed to decompose large-sized convolutional kernels into depthwise convolutions and pointwise convolutions. For M traditional convolutional kernels of size $K \times K$ and D dimensions, the parameter count is $K \times K \times M \times D$. Through depthwise separable convolution, the parameters can be reduced to $K \times K \times D + D \times M$, leading to a decrease in model parameters, improved computational efficiency, and maintained model performance.

The extracted feature maps are then stacked together, enabling the model to adaptively select between different kernel sizes depending on the specific task requirements. And then a local spatial attention model, as Eq. 2 is used to strengthen the features, which will provide a richer feature representation.

$$LSAM(F) = \sigma(\text{Conv}[\text{AvgPool}(F), \text{MaxPool}(F)]) \cdot F \quad (2)$$

The Transformer branch follows the architecture of ViT [26]. It partitions the input image into non-overlapping image patches of specified sizes, reshaping each patch into one-dimensional vectors (tokens). Subsequently, three matrices W_Q , W_K , and W_V are multiplied with the input vectors to obtain feature matrices Q , K , and V , respectively.

$$\begin{cases} Q = F_i W_Q \\ K = F_i W_K \\ V = F_i W_V \end{cases} \quad (3)$$

Subsequently, the expressive power of features is enhanced through the mechanism of self-attention, as depicted by the following equation:

$$\text{Atten}(Q, K, V) = \mathcal{S}(QK^T)V \quad (4)$$

In the equation, the dot product between Q and K facilitates the interaction of features among different patches, yielding the correlation between each pair of patches. $\mathcal{S}$ denotes the softmax function, which is applied along the rows to obtain attention weights. Subsequently, this result is multiplied with V , enabling the final features to automatically aggregate important information related to the target.

It should be noted that the computational complexity of Eq. 5 is proportional to the square of the token dimension. Therefore, when dealing with large input dimensions, the computational burden is considerable. However, expanding QK^T yields:

$$QK^T = F_i W_Q * W_K^T F_k^T = F_i W_{qk} F_k^T \quad (5)$$

Therefore, by merging W_K and W_Q , and using a single parameter matrix to map input vectors, one multiplication operation can be reduced, simultaneously decreasing the model's parameter count.

Thus, Eq. 5 can be rewritten as:

$$\text{Atten}(Q, K, V) = \text{Atten}(F_i, K', V) \quad (6)$$

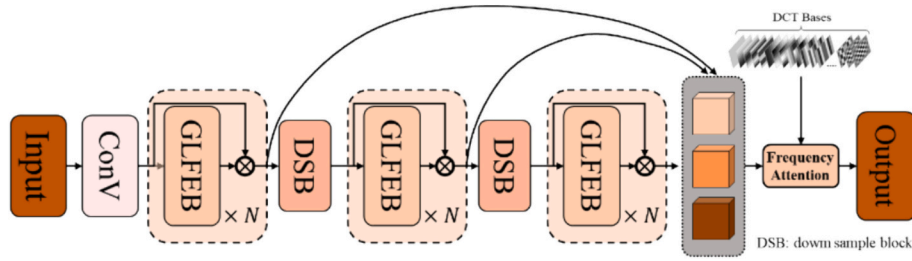


Fig. 3. The architecture of designed SSGLNet.

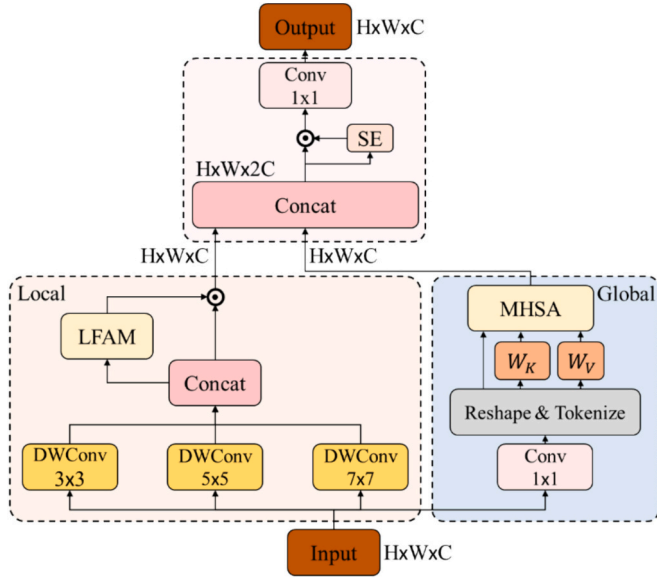


Fig. 4. The architecture of global-local feature extraction block.

where, $K' = F_i W_{qk}^T$.

Finally, a feature interaction block is employed to seamlessly integrate local features from CNN branch with global features from transformer branch. The SE (Squeeze-and-Excitation) structure [27] is utilized to establish connections between them. The local and global features are stacked together, and global average pooling is applied to aggregate information from each channel. Subsequently, two fully connected layers are utilized to obtain the importance weights of each channel feature through nonlinear mapping. Finally, these weights are applied to the corresponding channels, enabling the model to adaptively select and emphasize important features, thereby enhancing the global perception ability of local features and reinforcing the local details of global features.

3.1.2. Feature fusion block

As mentioned earlier, the model concatenates multiple layers of feature extraction blocks to capture image features at different hierarchical levels. As the concatenation depth increases, the expressive capability of features progressively strengthens, albeit at the expense of losing finer details at lower levels. This phenomenon is manifested in the spectrum by the prevalence of low-frequency signals in deep features and high-frequency signals in shallow features [28]. Therefore, a feature fusion block is devised to integrate feature maps from different depths., ensuring the representational capability and robustness of the output features.

A frequency domain attention block is established to compute the energy concentration of feature maps in different frequency bands using Discrete Cosine Transform (DCT) basis functions [29], as Eq. 2.

Subsequently, specific frequency band feature maps' contributions during model inference are enhanced through fully connected layers, facilitating frequency-domain attention among feature maps from different layers to improve model performance.

$$D = \cos\left(\frac{\pi h}{H}(i + 0.5)\right) \cos\left(\frac{\pi w}{W}(j + 0.5)\right) \quad (7)$$

By integrating semantic information from different frequency levels across various layers of feature maps ensures the diversity of feature spectra. Consequently, richer texture information and more accurate global information are obtained, thereby enhancing the discriminative ability of the features in subsequent tracking tasks.

3.2. Self-supervised learning based model training

As previously discussed, visual tracking tasks require detailed features to enhance object tracking accuracy while simultaneously leveraging global features to improve noise discernibility. Features extracted by deep network models have demonstrated excellent robustness in complex welding noise. However, their training process heavily relies on extensive data, and the laborious data collection and annotation tasks limit the size of the training samples, thereby somewhat diminishing the model's adaptability to complex welding scenarios (Fig. 5).

Hence, a jigsaw puzzle auxiliary task is proposed to guide SSGLNet's attention to common image features within a vast amount of unlabeled data, thereby enhancing the model's adaptability to complex welding scenes and reducing its reliance on sample labels during training as shown in Fig. 5.

Let the unlabeled dataset be denoted as $\mathcal{X} = \{x_u\}_{u=1}^{N_u}$, and denote the feature extraction network as $\mathcal{F}_M(\cdot|\theta)$. In each training batch, n images are selected and segmented into image blocks, which are then processed by $\mathcal{F}_M$ to obtain $n \times m \times m$ feature maps of scale c . Finally, two branches are added: clustering branch and localization branch. The clustering branch classifies all patches into n classes, grouping patches belonging to

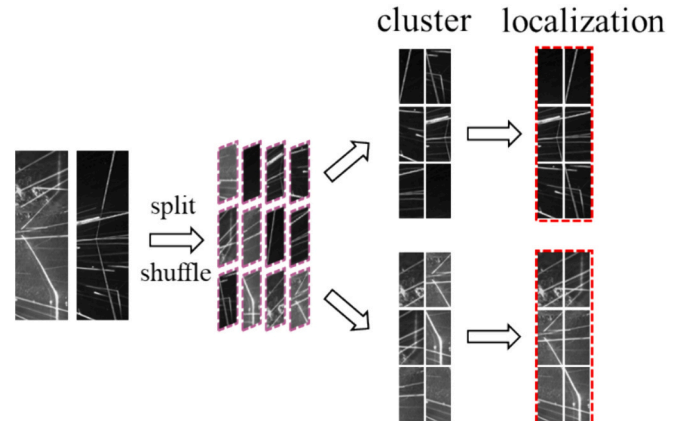


Fig. 5. The structure of the self-supervised training strategy.

the same image into one class. The localization branch is used to predict the positions of image blocks in the original image. The predictors for clustering prediction and position prediction tasks are denoted as $\mathcal{T}_R(\cdot|\vartheta_R)$ and $\mathcal{T}_P(\cdot|\vartheta_P)$ respectively, where ϑ_R and ϑ_P are trainable parameters in the two predictors. The training process is specifically described as follows.

In the clustering branch, inspired by contrastive learning principles, cosine similarity is used to measure the similarity between image blocks mapped by $\mathcal{T}_R$. Patches from the same image are clustered together, while patches from different images are separated. The following loss function calculates the error for the i -th and j -th patches belonging to the same class.

$$\ell_{ij} = -\log \frac{\exp(\cos(f_i, f_j)/\tau)}{\sum_{k=1, k \neq i}^{nm} \exp(\cos(f_i, f_k)/\tau)} \quad (8)$$

So, the overall loss for the clustering branch is the sum of losses for each patch, denoted as:

$$\mathcal{L}_{clu} = \frac{1}{nm} \sum_i \left(\frac{1}{mm-1} \sum_{j \in C_i} \ell_{ij} \right) \quad (9)$$

where, C_i represents the patches belonging to the same image as the i -th patch.

For the localization branch, we have defined it as a classification task. Let $\mathcal{T}_P(\cdot)_i$ represent the probability that the model predicts the input as the i -th type of variation. Using the cross-entropy function as the loss function, $\mathcal{L}_{pos}$ can be written as:

$$\mathcal{L}_{pos}(\mathbf{x}_u, \vartheta, \vartheta_R) = -\frac{1}{nm} \sum_{i=1}^{nm} \log(\mathcal{T}_P(\mathcal{T}_M(\mathcal{R}(\mathbf{x}_u|i)|\vartheta)|\vartheta_R)_i) \quad (10)$$

Ultimately, the model's self-supervised training process can be achieved by optimizing the following objective function:

$$\text{Loss} = \min_{\vartheta, \vartheta_R, \vartheta_P} \frac{1}{N} \sum_{u=1}^{N_u} (\mathcal{L}_{Rot}(\mathbf{x}_u, \vartheta, \vartheta_R) + \lambda \mathcal{L}_P(\mathbf{x}_u, \vartheta, \vartheta_P)) \quad (11)$$

where, λ is utilized to balance the contributions of $\mathcal{L}_{Rot}$ and $\mathcal{L}_P$ in the Loss.

In the task of image block clustering, the network requires a deep understanding of semantic information, such as spatial context within the images, to potentially group similar images together. Conversely, in localization tasks, the determination of image block positions often relies on certain detailed features within the images, such as edges and textures. Therefore, during the training process, $\mathcal{L}_{Rot}$ can guide SSGLNet to focus on global semantic information such as stripe distribution features within the images, while $\mathcal{L}_P$ can enhance SSGLNet's sensitivity to detailed object features within the images. By encouraging pay attention to these common within the laser stripe images through $\mathcal{L}_{Rot}$ and $\mathcal{L}_P$, it is advantageous in improving its performance in subsequent tracking tasks.

Moreover, to augment the complexity of model learning, random gaps are introduced during image block segmentation to prevent premature convergence induced by factors like edge continuity and pixel distribution. Furthermore, data augmentation techniques such as random pixel value perturbation and the addition of random noise are employed to bolster sample diversity and enhance the model's generalization capacity.

It should be noted that the self-supervised training process aims to guide the model to learn features relevant to subsequent tasks and typically does not focus on its performance in the pre-training task. Consequently, the more challenging the training task, the better the representation capability of the learned features and the transferability to downstream tasks.

4. Robust KCF tracker

As mentioned earlier, the tracker based key point tracking methods is suitable for diverse types of weld seams. In previous section we devised a deep neural network to ensure tracking performance at the feature level. Then, a robust KCF tracker is proposed in this section, which simultaneously ensures tracking efficiency and robustness against welding noise.

4.1. Brief review of KCF tracker

KCF (Kernelized Correlation Filters) is one of the most efficient object tracking frameworks currently available [30]. It trains a filter based on the target determined from the initial frame. Subsequently, using the principle of correlation filtering, it searches for the position most similar to the filter in subsequent frames. The maximum value in the response map is considered the new position of the target.

Generally, the tracking filter is defined as:

$$f(\mathbf{z}) = \omega^T \mathbf{z} \quad (12)$$

A Gaussian response map is generated around the target area, and then the filter is solved through least squares to ensure that the response map after filtering has a higher response in the vicinity of the target position. This process can be expressed as Eq. 14:

$$\min \sum_i (f(\mathbf{x}_i) - \mathbf{y}_i)^2 + \lambda \|\omega\|^2 \quad (13)$$

where, $\mathbf{y}_i$ is the output response map corresponding to the target $\mathbf{x}_i$. And λ is the regularization parameter to prevent overfitting.

Then, the parameters ω can be solved through ridge regression:

$$\omega = (X^T X + \lambda I)^{-1} X^T Y \quad (14)$$

where X and Y are respectively the unions of $\mathbf{x}_i$ and $\mathbf{y}_i$.

The primary contribution of the KCF algorithm is its utilization of circular matrix approach to expand the quantity of negative samples, thereby enhancing the performance of the tracker. It successfully leverages the property of circular matrices to diagonalize in Fourier space, thereby transforming matrix operations into vector Hadamard products, significantly reducing computational complexity and improving processing speed. The process of expanding the sample $\mathbf{x}_i$ through circular matrix operations $C(\cdot)$, as Fig. 6, to produce the augmented sample χ_i can be represented by the following equation:

$$\chi_i = C(\mathbf{x}_i) = F \bullet \text{diag}(\hat{\mathbf{x}}_i) \bullet F^H \quad (15)$$

where F represents the Fourier transform, satisfies $F \bullet \mathbf{x}_i = \hat{\mathbf{x}}_i$ and $F \bullet F^H = I$. And $\hat{\mathbf{x}}_i$ represents the Fourier transform of $\mathbf{x}_i$ (Fig. 6).

For one feature $\mathbf{x}_i$, according to the properties of the circular matrix, Eq. 15 can be simplified as:

$$\hat{\omega} = \frac{\hat{\mathbf{x}}_i \odot \hat{\mathbf{y}}}{\hat{\mathbf{x}}_i^* \odot \hat{\mathbf{x}}_i + \lambda} \quad (16)$$

Here, the superscript $\hat{\cdot}$ denotes the representation of parameters in the frequency domain after Fourier transformation. $*$ represents conjugation. $\odot$ denotes element-wise matrix multiplication.

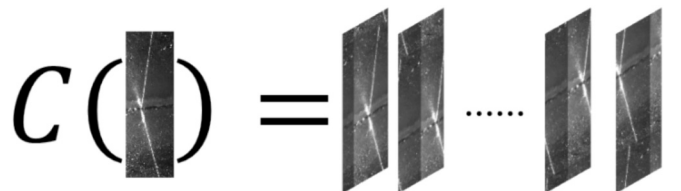


Fig. 6. Schematic diagram of the circular matrix operations $C(\cdot)$.

However, in most cases, $\hat{\omega}$ does not have a linear solution. Therefore, a kernel function is needed to map samples to a high-dimensional space. In this case:

$$\omega = \sum \alpha_i \varphi(\mathbf{x}_i) \quad (17)$$

Correspondingly, the solution for α in frequency domain is:

$$\hat{\alpha} = \frac{\hat{\mathbf{y}}}{\hat{\kappa}_{xx'} + \lambda} \quad (18)$$

where, $\hat{\kappa}_{xx'} = \varphi(\mathbf{x})\varphi^T(\mathbf{x}')$. For the commonly used Gaussian kernel, it is defined as:

$$\kappa_{xx'} = \exp\left(\|\mathbf{x}\| + \|\mathbf{x}'\| - 2\mathcal{F}^{-1}\left(\sum_c \hat{\mathbf{x}}_c^* \odot \hat{\mathbf{x}}_c'\right)\right) \quad (19)$$

When detecting targets, the response map can be computed as:

$$\hat{\mathbf{f}}(\mathbf{z}) = \hat{\kappa}_{xz} \odot \hat{\alpha} \quad (20)$$

It can be observed that in the process of solving the filter and generating the response map, the calculations mainly involve element-wise multiplication, significantly enhancing the efficiency of the algorithm.

4.2. Robust KCF based on multi-template enhanced strategy

In the traditional KCF algorithm, the detector is only generated from the initial frame. However, laser stripe images captured during welding processes often contain disturbances such as arc flashes, smoke, and splatters. A detector generated solely from clean initial frames lacks the discriminative ability to such noise. Therefore, introducing images with welding noise during the training process as training samples is beneficial for enhancing the performance of the obtained detector. This section will discuss such multi-template based tracking filter generation method in detail.

In the case of multiple templates, assuming the input sequence is $\mathcal{X} = [\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_m]$, where $\mathcal{X}_j$ represents the input corresponding to the j -th sample. Therefore, Eq. 14 can be rewritten as:

$$\min \sum_j \sum_i^n \left(f(\mathbf{x}_{ij}) - \mathbf{y}_{ij} \right)^2 + \lambda \|\omega\|^2 \quad (21)$$

In general, due to the different feature maps corresponding to different samples, i.e. $\mathbf{y}_{ij_1} \neq \mathbf{y}_{ij_2}$, it's challenging to obtain a closed-form solution for the above equation in nonlinear cases. This implies that traditional KCF algorithms can only utilize a single template to generate the tracker, which is not suitable for multi-template scenarios.

To address this issue, we simulate the generation of actual welding images by overlaying welding noise images on the initial stripe images to enhance the samples. In this case, the label sequences corresponding to different samples are the same, namely:

$$\mathbf{y}_{ij_1} = \mathbf{y}_{ij_2}, j_1, j_2 \in [1, m] \quad (22)$$

Moreover, it satisfies the relationship: $F \bullet \mathbf{y}_{j_1} \bullet F^H = F \bullet \mathbf{y}_{j_2} \bullet F^H$.

Consequently, Eq. 22 can be rewritten as:

$$\min \sum_j^m \left(\alpha \varphi^T(\mathcal{X}_j) * \varphi(\mathcal{X}_j) - \mathbf{y} \right)^2 + \frac{\lambda}{m} \sum_j^m \|\alpha \varphi^T(\mathcal{X}_j)\|^2 \quad (23)$$

Therefore, in the case of nonlinearity, the solution corresponding to Eq. 24 is:

$$\alpha = \frac{m\mathbf{y}}{\sum_j^m K_j + \lambda} \quad (24)$$

where, $K_j = \varphi(\mathcal{X}_j) * \varphi^T(\mathcal{X}_j)$.

By combining the properties of the circulant matrix: $C(K_j) = F \bullet \text{diag}(\hat{\kappa}_{xx}^j) \bullet F^H$, α can be transformed to frequency domain as:

$$\hat{\alpha} = \frac{m\hat{\mathbf{y}}}{\sum_j \hat{\kappa}_{xx}^j + \lambda} \quad (25)$$

Thus, the enhanced samples obtained by superimposing different welding noises can be used simultaneously to generate trackers for the same target. And comparing Eq. 19 and Eq. 26, it can be observed that Eq. 26 utilizes multiple samples to generate trackers, without significantly increasing the algorithm's computational cost, which helps improve the algorithm's discriminative ability towards stripes and noise. This enhancement strengthens its performance during the tracking process.

4.3. Evaluation of tracking results

During the tracking process, the predicted position of the target is determined by the peak position in the response map. However, when the response map is heavily disturbed by noise, resulting in false response peaks, it can lead to the failure of detecting weld seam key points. Therefore, real-time supervision of peak prediction results is necessary during the tracking process to ensure the accuracy of the algorithm's output. Utilizing the statistical regularity of response distribution near the maximum response peak to evaluate the confidence of tracking results is an effective approach.

The Peak to Sidelobe Ratio (PSR) is a commonly used metric for evaluating the effectiveness of correlation filter tracking. It evaluates the tracking results based on the response distribution in a certain neighborhood around the peak. It is defined as:

$$PSR = \frac{\mathcal{R}_{\max} - \mathcal{R}_{\text{mean}}}{\mathcal{R}_{\text{var}}} \quad (26)$$

where $\mathcal{R}_{\text{mean}}$ and $\mathcal{R}_{\text{var}}$ represent the mean and variance within the neighborhood of the peak value $\mathcal{R}_{\max}$, respectively.

PSR describes the prominence of $\mathcal{R}_{\max}$ in its neighborhood and reflects the strength of $\mathcal{R}_{\max}$. During the tracking process, when the target in the image being detected undergoes little change, such as disappearance, occlusion, motion blur, or similar interference, the response map conforms to a Gaussian distribution, resulting in a large PSR value. Conversely, when arc light locally obstructs the stripes or when splashes produce similar targets, the PSR value decreases. Therefore, the PSR value can describe the degree of match between the detected target and the template, with a higher PSR value indicating a more reliable detection result.

Considering the complex noise interference during the tracking process, a single indicator is insufficient to accurately reflect the reliability of the tracking results. Therefore, we introduce an additional indicator: Average Peak Correlation Energy (APCE) to evaluate the tracking results, defined as:

$$APCE = \frac{|\mathcal{R}_{\max} - \mathcal{R}_{\min}|}{\text{mean}(\sum_{w,h} |\mathcal{R}_{w,h} - \mathcal{R}_{\min}|)} \quad (27)$$

where $\mathcal{R}_{\min}$ is the minimum response value in the neighborhood, and $\mathcal{R}_{w,h}$ is the response at the coordinate (w, h) . APCE reflects the fluctuation level of the response map. When APCE undergoes significant changes over a certain period of time, it can be inferred that the tracker is contaminated or even lost.

In summary, this paper ultimately adopts the following formula to evaluate the tracking results:

$$\begin{cases} \frac{1}{T} \sum_{t=1}^T PSR_t < PSR_{\text{threshold}} \\ \frac{1}{T} \sum_{t=1}^T APCE_t < APCE_{\text{threshold}} \end{cases} \quad (28)$$

where $PSR_{threshold}$ represents the threshold for the PSR, and $APCE_{threshold}$ denotes the threshold for the APCE. Only when the mean of PSR and APCE over a period of time are both greater than the specified thresholds, the results are considered reliable. And this result will be used to update the tracking template using Eq. 30, allowing the model to adapt to changes in the welding process.

$$\hat{\alpha}_{new} = (1 - \tau)\hat{\alpha}_{old} + \tau\hat{\alpha}_{now} \quad (29)$$

where, τ is the learning rate, which controls the speed of model updates.

It should be noted that both APCE and PSR are metrics for tracking results, and they reflect the peak intensity in the response map and the target confidence using different statistical methods. According to the principle of independent probabilities, simultaneously evaluating tracking results using these two indicators can greatly enhance the reliability of the output results and help reduce the impact of welding noise on tracking stability.

5. Experimental validation and analysis

5.1. Experimental platform

5.1.1. Hardware platform

A laser vision sensor based welding seam tracking system is established to validate the effectiveness of the proposed algorithm, as shown in Fig. 7. The industrial computer has 8GB of memory and utilizes an Intel Core i7-9600 processor, along with a GTX 1060 GPU for processing operations. The seam tracking software running on the IPC can connect to the Fanuc robot via the TCP/IP protocol, utilizing the Robot Interface communication interface. The vision sensor is installed at the end of the welding torch, approximately 80 mm above the workpiece, with corresponding resolutions of 0.03 mm/pixel. The sampling frame rate is 60fps, and the original image size is 1200*1600. To expedite computations, a region of interest (ROI) sized 300*1000 is selected (Fig. 7).

5.1.2. Dataset preparation

To validate the effectiveness of the proposed algorithm, two datasets, D_U and D_L , were constructed using laser stripe images from daily welding experiments, as Fig. 8. D_U contains 12,000 images of various laser stripe types without any annotations, serving to validate the effectiveness of the feature extraction network based on self-supervised learning. D_L comprises sequential stripe images from different types of weld seams. These images are manually annotated to identify weld seam key point locations, serving as reference to validate the robustness and applicability of the tracking algorithm. Additionally, 2000 images are randomly selected from D_U , and laser stripe positions are annotated for

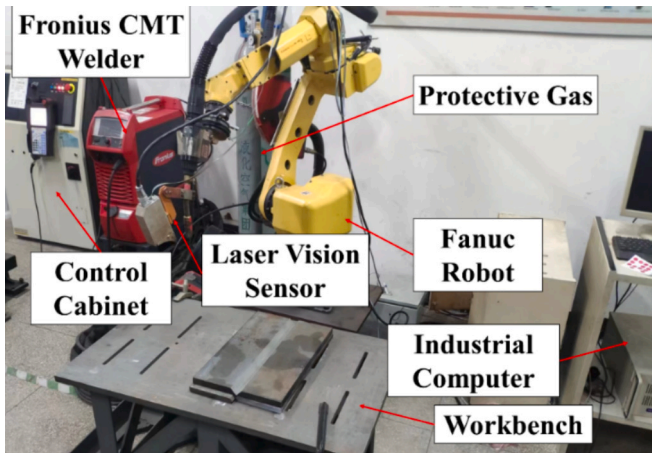


Fig. 7. Laser vision seam tracking experimental system.

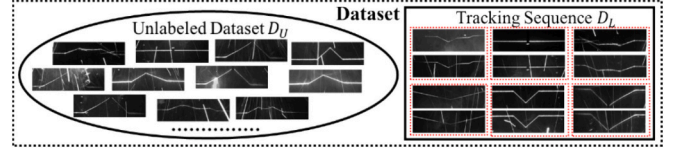


Fig. 8. Schematic diagram of the composition of the laser stripe dataset.

comparison with networks trained using traditional supervised learning methods, highlighting differences between them and the algorithm proposed in this paper (Fig. 8).

5.2. Model training and results

The feature extraction network was trained using the parameters listed in Table 1. A Warm-up training strategy was employed, where the a relatively low learning rate was used when starting training process, and then gradually increased it to the present value over the first 5 epochs. This approach will help stabilize the training process, and lead to faster convergence.

As shown in Fig. 9, the model attained high accuracy after 200 training epochs. As previously stated, given that the self-supervised training procedure aims solely to enhance the representational capacity of deep features for laser stripe images, achieving full convergence in training is not imperative. To prevent overfitting and ensure the model's optimal performance during tracking, training was terminated as accuracy approached 95 %.

After training, deep features corresponding to the positions of stripes and noise in 20 laser stripe images were obtained via the model. The visualization of feature vectors after dimensionality reduction using t-SNE is shown in the Fig. 10. It can be observed that the feature vectors of stripes and welding noise can be clearly distinguished, indicating that the network exhibits good discriminative ability towards welding noise and stripes.

Further examination of the convolutional kernels acquired through supervised and self-supervised training is depicted in Fig. 11. It can be observed that the shallow convolutional kernels obtained through supervised learning demonstrate a higher uniformity in weight distribution, leading to extracted features with greater similarity. In contrast, the convolutional kernels obtained through unsupervised learning exhibit greater diversity, ensuring a broader range of features and thereby enhancing the model's performance in subsequent tasks.

5.3. Algorithm performance testing

5.3.1. Algorithm adaptability testing

To validate the algorithm's adaptability to different types of welds under welding noise interference, this study evaluates its tracking performance on five kinds of laser stripe dataset from D_L , including V-groove joint (V), Butt joint (B), Fillet joints (T), and the second and third passes of multi-pass welding (MP2 and MP3 respectively).

The key point tracking errors are evaluated in pixel space as the Euclidean distance between the annotated locations p_i and the tracking results $\hat{p}_i$. And the mean and variance of the tracking errors are employed as evaluation metrics, as defined below:

Table 1
Model training parameter settings.

Parameters	Value
Batch size	32
Max epochs	300
Dropout keep probability	0.95
Weight decay	0.0001
The initial learning rate	0.005
The momentum value	0.98

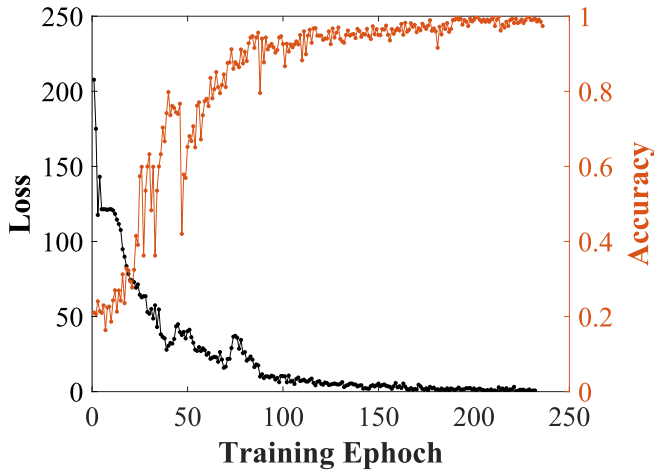


Fig. 9. The curve of training loss (left y-axis) and validation accuracy (right y-axis).

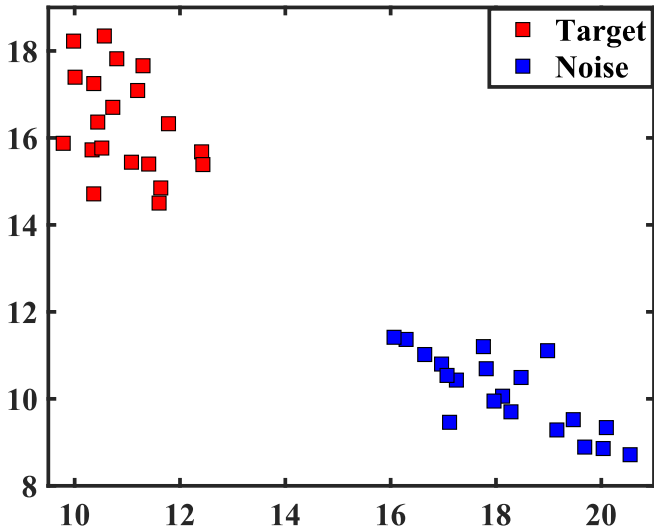


Fig. 10. The output features of stripes and noise were visualized with t-SNE.

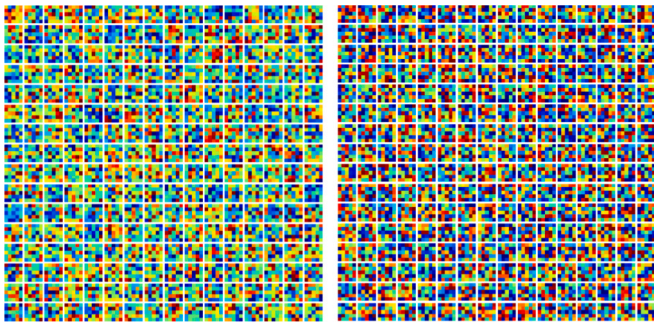


Fig. 11. The weight distribution of lower-level convolutional kernels trained by supervised (left) and self-supervised (right) methods.

$$\begin{cases} MAE = \frac{1}{N} \sum_{i=1}^N \|\hat{p}_i - p_i\| \\ Var = \frac{1}{N} \sum_{i=1}^N \|\hat{p}_i - MAE\| \end{cases} \quad (30)$$

Table 2 presents the experimental results, and the tracking results on V-groove joint are displayed in Fig. 12. It can be observed that the algorithm maintains an average tracking error of within 2 pixels across various types of weld seams, underscoring its high tracking accuracy. And the *Var* is less than 0.23, which demonstrates its tracking stability.

Throughout the self-supervised pre-training process, the model proficiently discerns the ubiquitous attributes of laser stripes from an extensive array of images, laying a solid foundation for accurately tracking. Meanwhile, the KCF tracker, reinforced by multi-template tricks, identifies the unique characteristics of the current tracking target, ensuring the stability of tracking processes across various weld seams. These enable the algorithm to maintain good tracking performance in scenarios with complex welding noise interference: In Fig. 13a) and b), despite splatter interference, some targets are partially obscured. In Fig. 13c), the image is affected by pulse welding, resulting in significant grayscale changes.

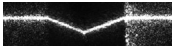
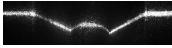
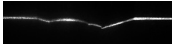
5.3.2. Ablation experiments

Ablation experiments were conducted to assess the impact of different modules on the tracking performance of the algorithm. Tracking errors, represented by *MAE* and *Var*, were evaluated across five categories of weld seams, totaling 2400 images, as detailed in Table 3. Here, M1 denotes the algorithm proposed in this paper, while M2 represents the KCF algorithm. Building upon M2, M3 was improved by incorporating multi-templates, whereas M4 and M5 utilized features from the proposed network and ResNet, respectively. To ensure the effectiveness of ResNet during tracking, it was pretrained in the segmentation task before use, following a similar approach to traditional algorithms [19,22].

By comparing M4, M5, and M2, it's evident that the utilization of deep features substantially enhances the stability of the KCF algorithm during the tracking process. The carefully designed network structure in this paper, tailored specifically for the tracking task to preserve both global and local features, effectively enhances the tracking performance compared to traditional networks. Additionally, comparing M1 with M4 and M2 with M3 reveals that employing multi-templates to establish the tracker also contributes to improving tracking accuracy. Each template, generated by overlaying welding noise on the tracking target, strengthens target-related features by calculating the convolution kernel using Eq. 3, thereby enhancing the tracker's adaptability to welding noise.

Table 2

Tracking performance on five kinds of laser stripe dataset.

Welding type	Laser stripe image	MAE (pixel)	Var
V		1.1	0.14
B		1.4	0.17
T		1.6	0.15
MP2		1.8	0.23
MP3		1.5	0.21

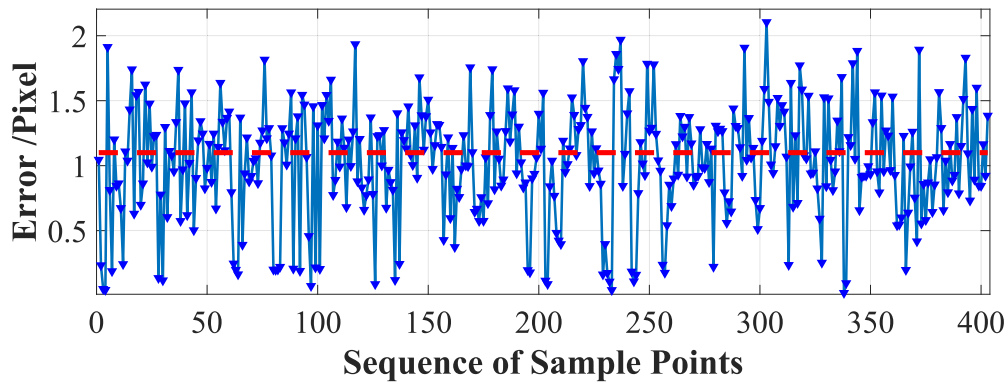


Fig. 12. Tracking accuracy on V-groove joint with different welding noise interference.

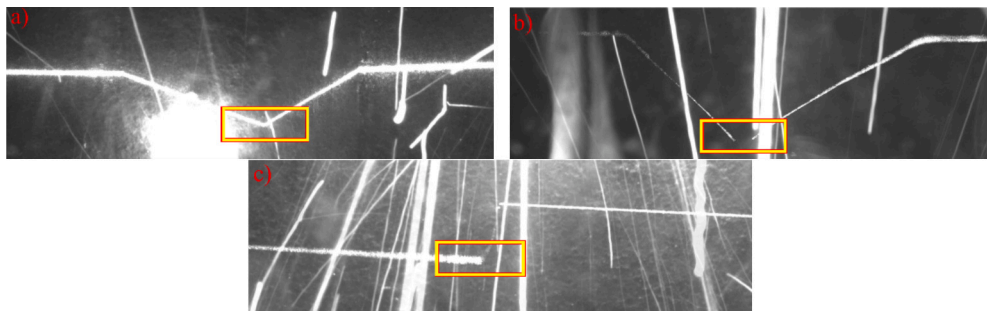


Fig. 13. Key point tracking results in special cases. The red and blue rectangles are labels and tracking results, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3

Tracking performance of different models.

Models	M1	M2	M3	M4	M5
MAE (pixel)	1.4	13.4	5.3	2.3	3.1
Var	0.18	1.4	0.73	0.31	0.45

5.3.3. Actual tracking experiments

Finally, tracking experiments were conducted on V-groove butt weld seam to validate the actual tracking performance of the algorithm in the robot's coordinate space. Firstly, the spatial equation of the weld seam was estimated through teaching. Then, the deviation was added to the teaching trajectory, represented by the yellow line in Fig. 14a). And the workpiece was fix with clamps to prevent the error caused by thermal deformation. During tracking, the spatial coordinates of the welding seam were reconstructed with Eq. 1, and the welding trajectory was corrected accordingly, as the red line. Finally, the distance from the measured points to the seam equation was calculated as the tracking error.

Besides, the welding experiments were conducted using the Fronius CMT power source, and the welding parameters were listed in Table 4.

The welding results are shown in Fig. 14b), and the tracking error is Fig. 14c). It can be seen that the mean tracking error is 0.21 mm, and the max error is less than 0.4 mm. Besides, Table 5 list the tracking results in Lap joint and Fillet joint, the mean tracking errors are both less than 0.25 mm, which demonstrates the satisfied tracking accuracy of the algorithm.

In summary, the proposed algorithm is fundamentally tracker-based. Similar to other algorithms of its kind, it is theoretically capable of achieving substantial tracking performance when weld seam features are distinct and the groove morphology remains relatively stable.

5.4. Comparison experiments

Finally, comparative experiments were conducted to showcase the superiority of the proposed algorithm by comparing its tracking performance with traditional tracking algorithms. Similarly, their tracking performance on five kinds of laser stripe dataset from D_L , including V, B, T, MP2, and MP3, and the MAE and VAR of tracking errors were employed as evaluation metrics. The results are presented in Table 6.

It is evident that the algorithm proposed in this paper outperforms existing algorithms in both tracking accuracy and stability. The method of smoothing point sequences using state-space equations fails to



Fig. 14. a) Seam tracking experiment, b) The tracking result, c) Tracking error in actual welding process.

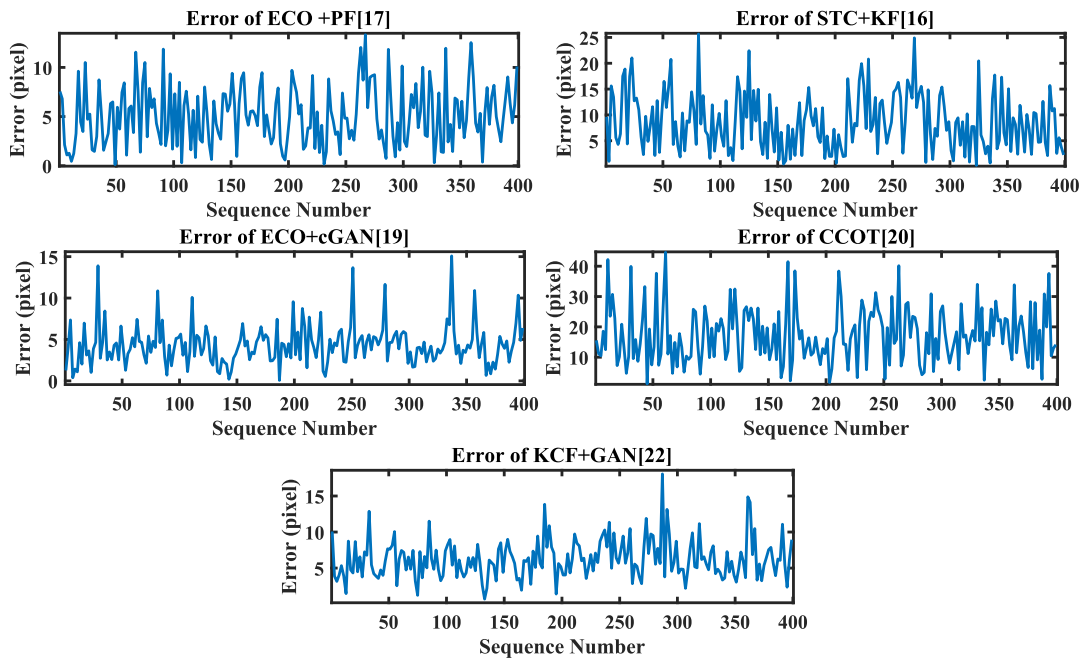


Fig. 15. Tracking results on V-groove welding seam.

Table 4
Welding parameters.

Parameters	Value	Parameters	Value
Material	Q235 steel plate	Welding speed	3 mm/s
Pulse frequency	125 HZ	Welding wire	1.2 mm
Welding current	140 A	Protective gas	80 % Ar + 20 % CO ₂
Dry elongation	15 mm	Gas-flow rate	14 L/min

fundamentally improve the tracker's resistance to weld seam noise, making it challenging to address complex noise interference. On the other hand, approaches using GANs to enhance image quality for improved tracking performance often require large network structures, resulting in lower computational efficiency. It is worth mentioning that the algorithm's utilization of deep features derived from self-supervised training enables it to capture common characteristics of stripes. This capability allows the algorithm to maintain high tracking performance even when tracking stripes that were not part of the training data.

Then, the efficiency of various algorithms was evaluated based on the average time required to process 2000 images. It can be observed that the proposed algorithm also demonstrates significant advantages in computational efficiency. This is attributed to the relatively low computational complexity of the KCF tracking process, and the multi-

template reinforcement strategy does not substantially increase the computational burden in tracking process. Moreover, the feature extraction network boasts a lightweight structure. With the combined effects of these factors, the algorithm significantly enhances tracking stability while ensuring real-time tracking with minimal computational cost.

Summarily, the proposed algorithm establishes a stripe feature extraction network from the perspective of model architecture and training strategy, aiming to enhance the model's ability to extract deep features for representing stripes effectively. Further, a multi-template reinforcement strategy is introduced to enhance the KCF tracking algorithm, thereby refining its ability to distinguish between tracking

Table 6
Algorithm performance comparison.

	MAE (pixel)	VAR	Time (ms)
ECO + PF [17]	5.1	0.29	11.4
STC + KF [16]	8.7	0.73	34.1
ECO + cGAN [19]	4.3	0.22	21.2
CCOT [20]	17.3	1.06	119.2
KCF + GAN [22]	6.2	0.54	18.3
Ours	1.4	0.18	6.5

Table 5
Tracking results on Lap and Fillet joints.

Seam type	Weld seam image	Welding current	Tracking error	MAE/mm
Lap		120 A		0.22
Fillet		180 A		0.23

targets and noise. The proposed algorithm demonstrates strong adaptability to various types of weld seams under welding noise interference while ensuring good real-time performance. Its advantages in both real-time performance and robustness make it a promising solution for weld seam tracking.

6. Conclusion

The key point tracking of weld seams based on the tracking algorithm has demonstrated remarkable performance in complex environments, demonstrating great potential for actual welding applications. To enhance the adaptability of the tracking algorithm to complex welding processes, this paper proposes a self-supervised robust KCF seam tracking algorithm. The following conclusions are drawn:

1) From the perspective of neural network architecture and training strategy, a self-supervised global-local feature extraction network (SSGLNet) is constructed. The model's structure is carefully designed to ensure its capability of extracting both global semantic information and local texture information. Additionally, a self-supervised training task is devised to guide the model to discern the common characteristics of stripes in unlabeled data. The features generated by SSGLNet possess strong representation capability for stripes, laying a solid foundation for the tracking process.

2) A multi-template training strategy is proposed to enhance the discriminative capability of the KCF tracker. The analytical solution for the multi-template enhanced KCF tracker is derived, which can improve tracker's tracking performance without significantly increasing its computational complexity.

3) The experimental results suggest that the proposed tracking algorithm is applicable to various types of weld seams and exhibits robustness to welding noise. The algorithm achieves an average tracking error of less than 0.25 mm and an average processing time of 6.5 ms, meeting the requirements for general seam tracking tasks. Compared to traditional algorithms, the proposed algorithm demonstrates advantages in terms of tracking accuracy, stability, and real-time performance.

Summarily, the combination of robust KCF and SSGLNet effectively enhances the algorithm's tracking performance under complex welding noise interference. And the algorithm depends minimally on labeled data when constructing the tracker, which make it a promising candidate method for realizing fully autonomous robotic welding.

CRedit authorship contribution statement

Runquan Xiao: Writing – original draft, Conceptualization. **Qixin Cao:** Supervision, Funding acquisition. **Shanben Chen:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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